

STAT 405 Project EDA

2026-03-11

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
# load in the 2018-2022 dataset
```

```
# the file was originally named "British Columbia_Full Data_data.csv"
```

```
# (for both the 2018-22 and 2020-24 datasets)
```

```
crash18_raw <- read.csv("icbc_crashes_2018_2022.csv", sep="\t", header=TRUE, fileEncoding="utf-16")
```

```
crash18 <- crash18_raw
```

```
# change the month from words to numbers
```

```
month_names <- c("JANUARY", "FEBRUARY", "MARCH", "APRIL", "MAY", "JUNE",  
                "JULY", "AUGUST", "SEPTEMBER", "OCTOBER", "NOVEMBER", "DECEMBER")
```

```
crash18$month <- match(crash18$Month.Of.Year, month_names)
```

```
crash18$year <- crash18$Date.Of.Loss.Year + (crash18$month - 1) / 12
```

```
head(crash18)
```

```
## Crash.Breakdown.2 Date.Of.Loss.Year Animal.Involved Crash.Severity
## 1 SURREY 2018 No CASUALTY CRASH
## 2 SURREY 2018 No CASUALTY CRASH
## 3 SURREY 2018 No CASUALTY CRASH
## 4 COQUITLAM 2018 No CASUALTY CRASH
## 5 SURREY 2018 No CASUALTY CRASH
## 6 PORT COQUITLAM 2018 No CASUALTY CRASH
## Cyclist.Involved Day.Of.Week Derived.Crash.Congifuration Heavy.Veh.Involved
## 1 No FRIDAY REAR END No
## 2 No FRIDAY REAR END No
## 3 No FRIDAY CONFLICTED No
## 4 No FRIDAY REAR END No
```

```

## 5          No          FRIDAY          REAR END          No
## 6          No          FRIDAY          SIDE IMPACT        No
## Intersection.Crash Month.Of.Year Motorcycle.Involved
## 1          No          SEPTEMBER          No
## 2          No          JUNE          No
## 3          No          APRIL          No
## 4          No          JANUARY          No
## 5          No          JUNE          No
## 6          No          SEPTEMBER          No
## Municipality.Name..ifnull. Parked.Vehicle.Involved Parking.Lot.Crash
## 1          SURREY          No          No
## 2          SURREY          No          No
## 3          SURREY          No          No
## 4          COQUITLAM          No          No
## 5          SURREY          No          No
## 6          PORT COQUITLAM          No          No
## Pedestrian.Involved          Region Street.Full.Name..ifnull. Time.Category
## 1          No LOWER MAINLAND          PORT MANN BRIDGE          06:00-08:59
## 2          No LOWER MAINLAND          PORT MANN BRIDGE          09:00-11:59
## 3          No LOWER MAINLAND          PORT MANN BRIDGE          12:00-14:59
## 4          No LOWER MAINLAND          GLEN DR          12:00-14:59
## 5          No LOWER MAINLAND          PORT MANN BRIDGE          12:00-14:59
## 6          No LOWER MAINLAND          BROADWAY ST          12:00-14:59
## Municipality.Name Road.Location.Description Street.Full.Name Metric.Selector
## 1          SURREY          PORT MANN BRIDGE PORT MANN BRIDGE          1
## 2          SURREY          PORT MANN BRIDGE PORT MANN BRIDGE          1
## 3          SURREY          PORT MANN BRIDGE PORT MANN BRIDGE          1
## 4          COQUITLAM          GLEN DR          GLEN DR          1
## 5          SURREY          PORT MANN BRIDGE PORT MANN BRIDGE          1
## 6          PORT COQUITLAM          BROADWAY ST          BROADWAY ST          1
## Total.Crashes Total.Victims Cross.Street.Full.Name Latitude Longitude
## 1          1          1          49.22589 -122.8180
## 2          1          1          49.22589 -122.8180
## 3          1          1          49.22589 -122.8180
## 4          1          1          49.28274 -122.8063
## 5          1          2          49.22589 -122.8180
## 6          1          1          49.24319 -122.7623
## Mid.Block.Crash Municipality.With.Boundary month          year
## 1          N          COQUITLAM & SURREY          9 2018.667
## 2          N          COQUITLAM & SURREY          6 2018.417
## 3          N          COQUITLAM & SURREY          4 2018.250
## 4          N          COQUITLAM          1 2018.000
## 5          N          COQUITLAM & SURREY          6 2018.417
## 6          N          PORT COQUITLAM          9 2018.667

```

```

# view crashes only in the City of Vancouver

crash18_cov <- filter(crash18, Crash.Breakdown.2 == "VANCOUVER")

head(crash18_cov)

```

```

## Crash.Breakdown.2 Date.Of.Loss.Year Animal.Involved Crash.Severity
## 1          VANCOUVER          2018          No CASUALTY CRASH
## 2          VANCOUVER          2018          No CASUALTY CRASH

```

## 3	VANCOUVER	2018	No CASUALTY CRASH
## 4	VANCOUVER	2018	No CASUALTY CRASH
## 5	VANCOUVER	2018	No CASUALTY CRASH
## 6	VANCOUVER	2018	No CASUALTY CRASH
##	Cyclist.Involved	Day.Of.Week	Derived.Crash.Congifuration Heavy.Veh.Involved
## 1	No	FRIDAY	SINGLE VEHICLE No
## 2	No	FRIDAY	UNDETERMINED No
## 3	No	FRIDAY	CONFLICTED No
## 4	No	FRIDAY	REAR END No
## 5	No	FRIDAY	REAR END No
## 6	No	FRIDAY	CONFLICTED No
##	Intersection.Crash	Month.Of.Year	Motorcycle.Involved
## 1	No	JANUARY	No
## 2	No	DECEMBER	No
## 3	No	SEPTEMBER	No
## 4	No	JUNE	No
## 5	No	JUNE	No
## 6	No	MAY	No
##	Municipality.Name..ifnull.	Parked.Vehicle.Involved	Parking.Lot.Crash
## 1	VANCOUVER	No	No
## 2	VANCOUVER	No	No
## 3	VANCOUVER	No	No
## 4	VANCOUVER	No	No
## 5	VANCOUVER	No	No
## 6	VANCOUVER	No	No
##	Pedestrian.Involved	Region	Street.Full.Name..ifnull. Time.Category
## 1	No	LOWER MAINLAND	TRANS-CANADA HWY OFFRAMP 00:00-02:59
## 2	No	LOWER MAINLAND	UNKNOWN 00:00-02:59
## 3	No	LOWER MAINLAND	Unknown 06:00-08:59
## 4	No	LOWER MAINLAND	W WATERFRONT RD 06:00-08:59
## 5	No	LOWER MAINLAND	UNKNOWN 06:00-08:59
## 6	No	LOWER MAINLAND	W 16TH AVE 09:00-11:59
##	Municipality.Name	Road.Location.Description	Street.Full.Name
## 1	VANCOUVER	UNKNOWN	TRANS-CANADA HWY OFFRAMP
## 2	VANCOUVER	UNKNOWN	UNKNOWN
## 3	VANCOUVER	UNKNOWN	
## 4	VANCOUVER	W WATERFRONT RD	W WATERFRONT RD
## 5	VANCOUVER	UNKNOWN	UNKNOWN
## 6	VANCOUVER	W 16TH AVE	W 16TH AVE
##	Metric.Selector	Total.Crashes	Total.Victims Cross.Street.Full.Name Latitude
## 1	1	1	1 BOUNDARY RD NA
## 2	1	1	1 NA
## 3	1	1	2 NA
## 4	1	1	1 NA
## 5	1	1	3 NA
## 6	1	1	1 NA
##	Longitude	Mid.Block.Crash	Municipality.With.Boundary month year
## 1	NA	N	1 2018.000
## 2	NA	N	12 2018.917
## 3	NA	N	9 2018.667
## 4	NA	N	6 2018.417
## 5	NA	N	6 2018.417
## 6	NA	N	5 2018.333

```
# load in the 2020-2024 crash dataset
crash20_raw <- read.csv("icbc_crashes_2020_2024.csv", header=TRUE)
crash20 <- crash20_raw
```

```
crash20$month <- match(crash20$Month.Of.Year, month_names)
crash20$year <- crash20$Date.Of.Loss.Year + (crash20$month - 1) / 12

head(crash20)
```

```
## Crash.Breakdown.2 Date.Of.Loss.Year Animal.Flag Crash.Severity Cyclist.Flag
## 1 DELTA 2020 No CASUALTY CRASH No
## 2 MISSION 2020 No CASUALTY CRASH No
## 3 ABBOTSFORD 2020 No CASUALTY CRASH No
## 4 COQUITLAM 2020 No CASUALTY CRASH No
## 5 MISSION 2020 No CASUALTY CRASH No
## 6 SURREY 2020 No CASUALTY CRASH No
## Day.Of.Week Derived.Crash.Configuration Heavy.Veh.Flag Intersection.Crash
## 1 FRIDAY REAR END No No
## 2 MONDAY CONFLICTED No No
## 3 MONDAY SINGLE VEHICLE No No
## 4 SATURDAY REAR END No No
## 5 SUNDAY SINGLE VEHICLE No No
## 6 THURSDAY CONFLICTED No No
## Month.Of.Year Motorcycle.Flag Municipality.Name..ifnull. Parked.Vehicle.Flag
## 1 JANUARY No DELTA No
## 2 JULY No MISSION No
## 3 DECEMBER No ABBOTSFORD No
## 4 SEPTEMBER No COQUITLAM No
## 5 NOVEMBER No MISSION No
## 6 APRIL No SURREY No
## Parking.Lot.Flag Pedestrian.Flag Region Road.Location.Description
## 1 No No LOWER MAINLAND UNKNOWN
## 2 No No LOWER MAINLAND UNKNOWN
## 3 No No LOWER MAINLAND UNKNOWN
## 4 No No LOWER MAINLAND UNKNOWN
## 5 No No LOWER MAINLAND UNKNOWN
## 6 No No LOWER MAINLAND UNKNOWN
## Street.Full.Name..ifnull. Time.Category Municipality.Name Street.Full.Name
## 1 HWY 99 15:00-17:59 DELTA HWY 99
## 2 MISSION BRIDGE 15:00-17:59 MISSION MISSION BRIDGE
## 3 VYE RD 18:00-20:59 ABBOTSFORD VYE RD
## 4 BRUNETTE AVE 12:00-14:59 COQUITLAM BRUNETTE AVE
## 5 FARMS RD 15:00-17:59 MISSION FARMS RD
## 6 HWY 1 09:00-11:59 SURREY HWY 1
## Metric.Selector Total.Crashes Total.Victims month year
## 1 1 1 1 1 2020.000
## 2 1 1 3 7 2020.500
## 3 1 1 2 12 2020.917
## 4 1 1 2 9 2020.667
## 5 1 1 2 11 2020.833
## 6 1 1 3 4 2020.250
```

```
# filter out 2020-2022 crashes so only 2023-2024 crashes remain
# so we can then combine the two datasets to have 2018-2024 data
```

```
crash23 <- crash20 |> filter(year >= 2023)
head(crash23)
```

```
##      Crash.Breakdown.2 Date.Of.Loss.Year Animal.Flag      Crash.Severity
## 1          SURREY          2023          No PROPERTY DAMAGE ONLY
## 2          VANCOUVER          2023          No PROPERTY DAMAGE ONLY
## 3          BURNABY          2023          No PROPERTY DAMAGE ONLY
## 4          VANCOUVER          2023          No PROPERTY DAMAGE ONLY
## 5 NORTH VANCOUVER          2023          No PROPERTY DAMAGE ONLY
## 6          MISSION          2023          No PROPERTY DAMAGE ONLY
##      Cyclist.Flag Day.Of.Week Derived.Crash.Configuration Heavy.Veh.Flag
## 1          No      FRIDAY          REAR END          No
## 2          No      FRIDAY          REAR END          No
## 3          No      FRIDAY          SIDE IMPACT          No
## 4          No      FRIDAY          SIDE IMPACT          No
## 5          No      FRIDAY          SIDE IMPACT          No
## 6          No      FRIDAY          SIDE IMPACT          No
##      Intersection.Crash Month.Of.Year Motorcycle.Flag Municipality.Name..ifnull.
## 1          Yes      SEPTEMBER          No          SURREY
## 2          Yes      SEPTEMBER          No          VANCOUVER
## 3          Yes      APRIL          No          BURNABY
## 4          Yes      APRIL          No          VANCOUVER
## 5          Yes      AUGUST          No          NORTH VANCOUVER
## 6          Yes      JANUARY          No          MISSION
##      Parked.Vehicle.Flag Parking.Lot.Flag Pedestrian.Flag      Region
## 1          No          No          No LOWER MAINLAND
## 2          No          No          No LOWER MAINLAND
## 3          No          No          No LOWER MAINLAND
## 4          No          No          No LOWER MAINLAND
## 5          No          No          No LOWER MAINLAND
## 6          No          No          No LOWER MAINLAND
##      Road.Location.Description Street.Full.Name..ifnull. Time.Category
## 1      64 AVE & FRASER HWY & TURNING LANE      64 AVE      09:00-11:59
## 2      GRANDVIEW HWY & SKEENA ST      GRANDVIEW HWY      09:00-11:59
## 3 BYRNE RD & MARINE WAY & TURNING LANE      MARINE WAY      09:00-11:59
## 4      CONNAUGHT DR & W 33RD AVE      W 33RD AVE      09:00-11:59
## 5      CHESTERFIELD AVE & W 16TH ST      CHESTERFIELD AVE      09:00-11:59
## 6      ORIOLE AVE & ROBIN CRES      ORIOLE AVE      09:00-11:59
##      Municipality.Name Street.Full.Name Metric.Selector Total.Crashes
## 1          SURREY      64 AVE          1          1
## 2          VANCOUVER      GRANDVIEW HWY          1          1
## 3          BURNABY      MARINE WAY          1          1
## 4          VANCOUVER      W 33RD AVE          1          1
## 5 NORTH VANCOUVER CHESTERFIELD AVE          1          1
## 6          MISSION      ORIOLE AVE          1          1
##      Total.Victims month      year
## 1          0          9 2023.667
## 2          0          9 2023.667
## 3          0          4 2023.250
## 4          0          4 2023.250
```

```
## 5          0      8 2023.583
## 6          0      1 2023.000
```

```
# rename columns in crash18 to match those in crash23
crash18 <- crash18|> rename(Animal.Flag = Animal.Involved,
                           Cyclist.Flag = Cyclist.Involved,
                           Heavy.Veh.Flag = Heavy.Veh.Involved,
                           Motorcycle.Flag = Motorcycle.Involved,
                           Parked.Vehicle.Flag = Parked.Vehicle.Involved,
                           Parking.Lot.Flag = Parking.Lot.Crash,
                           Pedestrian.Flag = Pedestrian.Involved)
```

```
# combine the crash18 and crash23 data frames
```

```
crash <- bind_rows(crash18, crash23)

head(crash)
```

```
##   Crash.Breakdown.2 Date.Of.Loss.Year Animal.Flag Crash.Severity Cyclist.Flag
## 1          SURREY          2018          No CASUALTY CRASH          No
## 2          SURREY          2018          No CASUALTY CRASH          No
## 3          SURREY          2018          No CASUALTY CRASH          No
## 4      COQUITLAM          2018          No CASUALTY CRASH          No
## 5          SURREY          2018          No CASUALTY CRASH          No
## 6  PORT COQUITLAM          2018          No CASUALTY CRASH          No
##   Day.Of.Week Derived.Crash.Congifuration Heavy.Veh.Flag Intersection.Crash
## 1    FRIDAY          REAR END          No          No
## 2    FRIDAY          REAR END          No          No
## 3    FRIDAY      CONFLICTED          No          No
## 4    FRIDAY          REAR END          No          No
## 5    FRIDAY          REAR END          No          No
## 6    FRIDAY      SIDE IMPACT          No          No
##   Month.Of.Year Motorcycle.Flag Municipality.Name..ifnull. Parked.Vehicle.Flag
## 1    SEPTEMBER          No          SURREY          No
## 2        JUNE          No          SURREY          No
## 3      APRIL          No          SURREY          No
## 4    JANUARY          No      COQUITLAM          No
## 5        JUNE          No          SURREY          No
## 6    SEPTEMBER          No  PORT COQUITLAM          No
##   Parking.Lot.Flag Pedestrian.Flag      Region Street.Full.Name..ifnull.
## 1          No          No LOWER MAINLAND  PORT MANN BRIDGE
## 2          No          No LOWER MAINLAND  PORT MANN BRIDGE
## 3          No          No LOWER MAINLAND  PORT MANN BRIDGE
## 4          No          No LOWER MAINLAND      GLEN DR
## 5          No          No LOWER MAINLAND  PORT MANN BRIDGE
## 6          No          No LOWER MAINLAND  BROADWAY ST
##   Time.Category Municipality.Name Road.Location.Description Street.Full.Name
## 1  06:00-08:59          SURREY  PORT MANN BRIDGE PORT MANN BRIDGE
## 2  09:00-11:59          SURREY  PORT MANN BRIDGE PORT MANN BRIDGE
## 3  12:00-14:59          SURREY  PORT MANN BRIDGE PORT MANN BRIDGE
## 4  12:00-14:59      COQUITLAM      GLEN DR      GLEN DR
## 5  12:00-14:59          SURREY  PORT MANN BRIDGE PORT MANN BRIDGE
## 6  12:00-14:59  PORT COQUITLAM  BROADWAY ST  BROADWAY ST
```

##	Metric.Selector	Total.Crashes	Total.Victims	Cross.Street.Full.Name	Latitude
## 1	1	1	1		49.22589
## 2	1	1	1		49.22589
## 3	1	1	1		49.22589
## 4	1	1	1		49.28274
## 5	1	1	2		49.22589
## 6	1	1	1		49.24319

##	Longitude	Mid.Block.Crash	Municipality.With.Boundary	month	year
## 1	-122.8180	N	COQUITLAM & SURREY	9	2018.667
## 2	-122.8180	N	COQUITLAM & SURREY	6	2018.417
## 3	-122.8180	N	COQUITLAM & SURREY	4	2018.250
## 4	-122.8063	N	COQUITLAM	1	2018.000
## 5	-122.8180	N	COQUITLAM & SURREY	6	2018.417
## 6	-122.7623	N	PORT COQUITLAM	9	2018.667

##	Derived.Crash.Configuration
## 1	<NA>
## 2	<NA>
## 3	<NA>
## 4	<NA>
## 5	<NA>
## 6	<NA>

```
tail(crash)
```

##	Crash.Breakdown.2	Date.Of.Loss.Year	Animal.Flag	Crash.Severity
## 1974713	RICHMOND	2023	No	CASUALTY CRASH
## 1974714	SURREY	2023	No	CASUALTY CRASH
## 1974715	SURREY	2023	No	CASUALTY CRASH
## 1974716	VANCOUVER	2023	No	CASUALTY CRASH
## 1974717	VANCOUVER	2023	No	CASUALTY CRASH
## 1974718	WEST VANCOUVER	2023	No	CASUALTY CRASH

##	Cyclist.Flag	Day.Of.Week	Derived.Crash.Congifuration	Heavy.Veh.Flag
## 1974713	No	WEDNESDAY	<NA>	No
## 1974714	No	WEDNESDAY	<NA>	No
## 1974715	No	WEDNESDAY	<NA>	No
## 1974716	No	WEDNESDAY	<NA>	No
## 1974717	No	WEDNESDAY	<NA>	No
## 1974718	No	WEDNESDAY	<NA>	No

##	Intersection.Crash	Month.Of.Year	Motorcycle.Flag
## 1974713	Yes	MARCH	No
## 1974714	Yes	MARCH	No
## 1974715	Yes	MARCH	No
## 1974716	Yes	MARCH	No
## 1974717	Yes	MARCH	No
## 1974718	Yes	MARCH	No

##	Municipality.Name..ifnull.	Parked.Vehicle.Flag	Parking.Lot.Flag
## 1974713	RICHMOND	No	No
## 1974714	SURREY	No	No
## 1974715	SURREY	No	No
## 1974716	VANCOUVER	No	No
## 1974717	VANCOUVER	No	No
## 1974718	WEST VANCOUVER	No	No

##	Pedestrian.Flag	Region	Street.Full.Name..ifnull.	Time.Category
## 1974713	No	LOWER MAINLAND	WESTMINSTER HWY	15:00-17:59

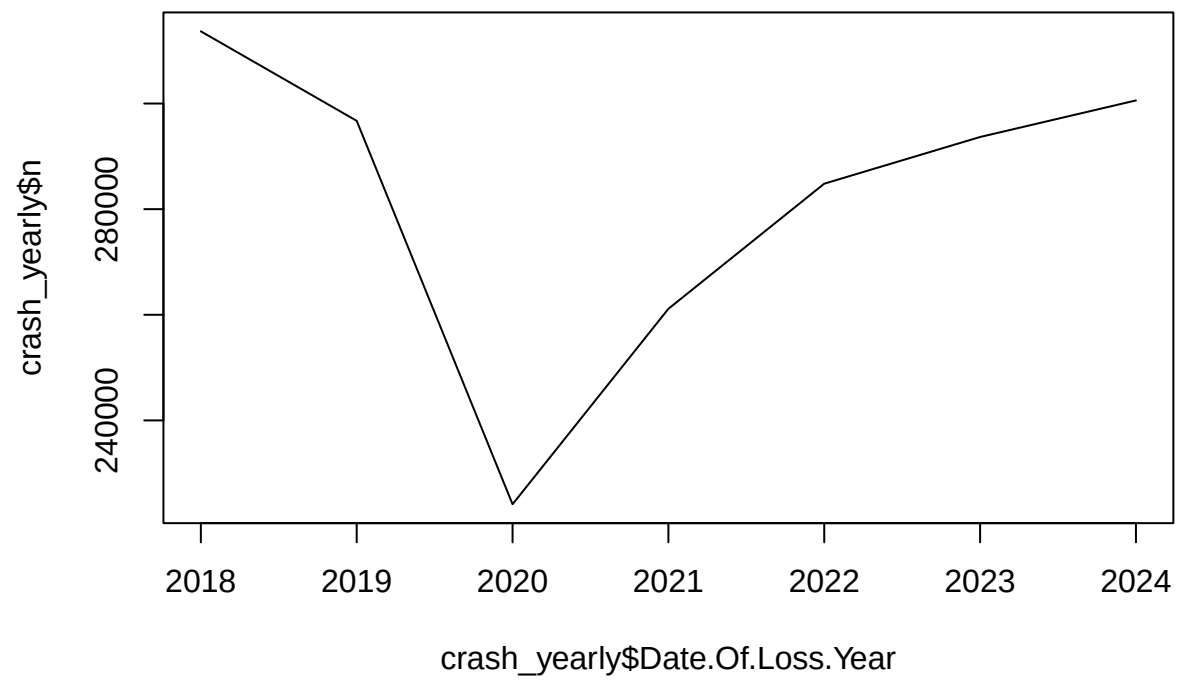
```
## 1974714      No LOWER MAINLAND      HWY 1  15:00-17:59
## 1974715      No LOWER MAINLAND      140 ST 15:00-17:59
## 1974716      No LOWER MAINLAND      FRASER ST 15:00-17:59
## 1974717      No LOWER MAINLAND      MAIN ST 15:00-17:59
## 1974718      No LOWER MAINLAND      MARINE DR 15:00-17:59
##      Municipality.Name      Road.Location.Description
## 1974713      RICHMOND      NO 5 RD & WESTMINSTER HWY & TURNING LANE
## 1974714      SURREY 152 ST OFFRAMP & 152 ST ONRAMP & TRANS-CANADA HWY
## 1974715      SURREY      140 ST & FRASER HWY
## 1974716      VANCOUVER      E 12TH AVE & FRASER ST
## 1974717      VANCOUVER      E KING EDWARD AVE & MAIN ST
## 1974718      WEST VANCOUVER      15TH ST & MARINE DR
##      Street.Full.Name Metric.Selector Total.Crashes Total.Victims
## 1974713      WESTMINSTER HWY      1      1      1
## 1974714      HWY 1      1      1      1
## 1974715      140 ST      1      1      1
## 1974716      FRASER ST      1      1      1
## 1974717      MAIN ST      1      1      1
## 1974718      MARINE DR      1      1      1
##      Cross.Street.Full.Name Latitude Longitude Mid.Block.Crash
## 1974713      <NA>      NA      NA      <NA>
## 1974714      <NA>      NA      NA      <NA>
## 1974715      <NA>      NA      NA      <NA>
## 1974716      <NA>      NA      NA      <NA>
## 1974717      <NA>      NA      NA      <NA>
## 1974718      <NA>      NA      NA      <NA>
##      Municipality.With.Boundary month      year Derived.Crash.Configuration
## 1974713      <NA>      3 2023.167      REAR END
## 1974714      <NA>      3 2023.167      REAR END
## 1974715      <NA>      3 2023.167      REAR END
## 1974716      <NA>      3 2023.167      REAR END
## 1974717      <NA>      3 2023.167      REAR END
## 1974718      <NA>      3 2023.167      REAR END
```

```
# crashes per year
crash_yearly <- crash |> count(Date.Of.Loss.Year)

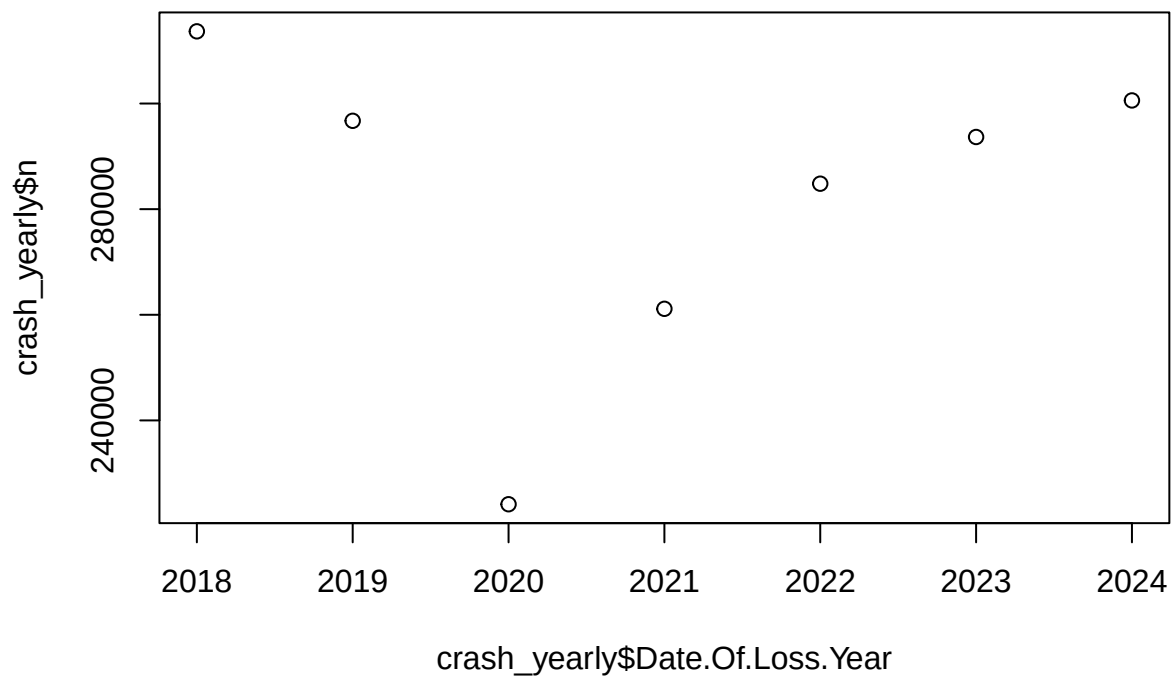
crash_yearly
```

```
##      Date.Of.Loss.Year      n
## 1      2018 313665
## 2      2019 296728
## 3      2020 224128
## 4      2021 261130
## 5      2022 284823
## 6      2023 293662
## 7      2024 300582
```

```
plot(crash_yearly$Date.Of.Loss.Year, crash_yearly$n, type="l")
```



```
plot(crash_yearly$Date.Of.Loss.Year, crash_yearly$n)
```

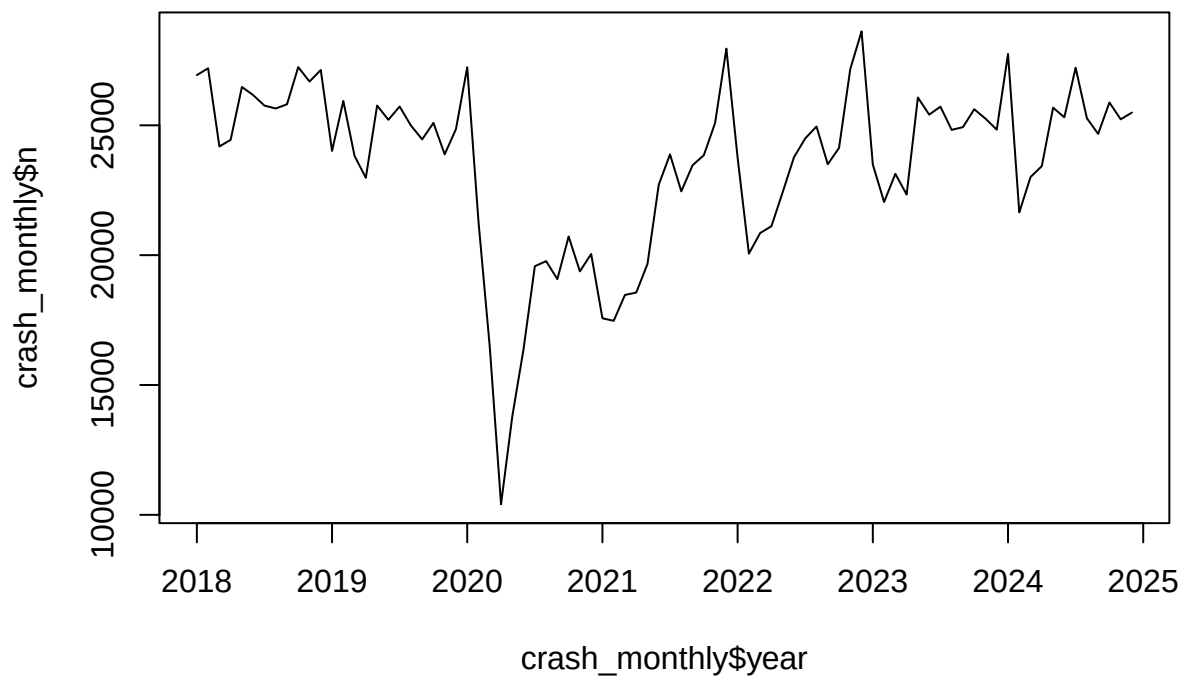


```
# crashes per month, to see if there is seasonality
crash_monthly <- crash |> count(year)
```

```
head(crash_monthly)
```

```
##      year      n
## 1 2018.000 26933
## 2 2018.083 27196
## 3 2018.167 24186
## 4 2018.250 24439
## 5 2018.333 26477
## 6 2018.417 26163
```

```
plot(crash_monthly$year, crash_monthly$n, type="l")
```



```
plot(crash_monthly$year, crash_monthly$n)
```

